Part I

Category theory

Chapter 1

Category Theory

Definition 1.0.1 (Category)**.** A **category** C consists of:

- A collection of *objects* $X, Y, \ldots$
- For each pair of objects X, Y a set of morphisms $\mathrm{Hom}_{\mathsf{C}}(X, Y)$. We denote this morphisms by $f : X \to Y$ or $X \xrightarrow{f} Y$.

such that

- Each object has an *identity morphism* $\mathbb{1}_X : X \to X$.
- For each pair of morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$ there exists a *composite morphism* $gf : X \to Z$.

We impose two axioms on this data

1. For any $f : X \to Y$ the composites $\mathbb{1}_Y f$ and $f\mathbb{1}_X$ are equal to f.
2. For any triplet of morphisms $X \xrightarrow{f} Y \xrightarrow{g} W \xrightarrow{h} Z$ the composites $h(gf)$ and $(hg)f$ are equal. We denote this composite by hgf.

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Remark 1.0.1. Technically speaking Definition 1.0.1 is the definition of a *locally small category*. In general, a category C can have an arbitrarily *large* collection of morphisms between two objects. ▽

Every Category has an associated opposite category

Definition 1.0.2 (Opposite Category)**.** Let C be a Category. The **opposite Category** of C, C^{op} has

- The same objects as C.

- For every pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.

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Definition 1.0.3. A morphism $f : X \to Y$ in a Category C is **monomorphism** if for any parallel pair of morphisms $h, k : W \rightrightarrows X, fh = fk$ implies $h = k$. Dually, f is an **epimorphism** if for any pair of morphisms $h, k : Y \rightrightarrows Z, hf = kf$ implies $h = k$. ⌟

Definition 1.0.4 (Isomorphism)**.** A morphism $f : x \to y$ in a Category C is an **isomorphism** if there exists a morphism $g : y \to x$ in C such that $fg = \mathbb{1}_y$ and $gf = \mathbb{1}_x$.

The objects x and y are called *isomorphic* if there exists an isomorphism between them, and we denote this by $x \cong y$. ⌟

Notation. We denote monomorphisms by $f : X \rightarrowtail Y$, epimorphisms by $f : X \twoheadrightarrow y$ and Isomorphisms by $f : X \xrightarrow{\sim} Y$.

Definition 1.0.5 (Functor)**.** A **functor** $F : \mathsf{C} \to \mathsf{D}$ between two Categories consists of:

- For each object c in C an object $Fc \in \mathsf{D}$.
- For each morphism $f : c \to c'$ in C a morphism $Ff : Fc \to Fc'$ in D.

satisfying the following axioms

1. For each element c in C, $F\mathbb{1}_c = \mathbb{1}_{Fc}$.
2. For any composable pair of morphisms f, g in C, $Fg \cdot Ff = F(g \cdot f)$.

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Notation. A Functor of the form $\mathsf{C} \to \mathsf{D}$ is usually called a *covariant functor* from C to D while a functor of the form $\mathsf{C}^{\mathrm{op}} \to \mathsf{D}$ is called a *contravariant* functor from C to D.

Example 1.0.1 (Posetal category of open sets)**.** The Functor $\mathsf{Op} : \mathsf{Top} \to \mathsf{Poset}$ sends a topological space X to its poset of open sets.

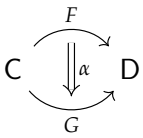
Therefore for every topological space with have an associated *posetal category of open sets* $\mathsf{Op}(X)$. ◊

Definition 1.0.6 (Natural transformation)**.** Let C, D be two Categories and $F, G : \mathsf{C} \rightrightarrows \mathsf{D}$ be two Functors. A **natural transformation** $\alpha : F \Rightarrow G$ consists of:

- For each object $c \in \mathsf{C}$ a morphism$\alpha_c : Fc \to Gc$ in D such that, for any morphism $f : c \to c'$ in C the following diagram commutes in D

$$\begin{array}{ccc} Fc & \xrightarrow{\alpha_c} & Gc \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ Fc' & \xrightarrow[\alpha_{c'}]{} & Gc' \end{array}$$

We denote this by



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Definition 1.0.7. A Natural transformation $\alpha : F \Rightarrow G$ is a **natural isomorphism** if every component α_f is an Isomorphism.

In this case, we say that F and G are **naturally isomorphic** and we denote it by $F \cong G$. ⌟

Definition 1.0.8. A Functor $F : \mathsf{C} \to \mathsf{D}$ is

- **full** if for all pair of objects $x, y \in \mathsf{C}$ the map $\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(Fx, Fy)$ is surjective.
- **faithful** if for all pair of objects $x, y \in \mathsf{C}$ the map $\mathrm{Hom}_{\mathsf{C}}(x, y) \to \mathrm{Hom}_{\mathsf{D}}(Fx, Fy)$ is injective.
- **essentially surjective on objects** if for every object $d \in \mathsf{D}$ there exists $c \in \mathsf{C}$ such that $d \cong Fc$.

If F is both full and faithfull we will say it is **fully faithful**. ⌟

Definition 1.0.9 (Concrete category)**.** A Category C is a **concrete Category** if there exists a faithul functor $F : \mathsf{C} \to \mathsf{Set}$. ⌟

Definition 1.0.10 (Diagram)**.** A **diagram** in a Category C with shape J is a Functor $F : \mathsf{C} \to \mathsf{J}$. ⌟

Definition 1.0.11 (Initial and terminal objects)**.** Let C be a Category. An object $i \in \mathsf{C}$ is **initial** if for every object $c \in \mathsf{C}$ there exists a unique morphism $i \to c$. Dually, an object $t \in \mathsf{C}$ is **terminal** if for every object $c \in \mathsf{C}$ there exists a unique morphism $c \to t$. ⌟

Definition 1.0.12 (Zero object)**.** An object $0 \in \mathsf{C}$ is a **zero object** if it is both an initial and terminal object. ⌟

Definition 1.0.13. Let c be an object in a Category C. The **Functor represented** by c is the Functor $\mathrm{Hom}_{\mathsf{C}}(c, -) \to \mathsf{Set}$ defined as

$$\begin{array}{ccc} x & & \mathrm{Hom}_{\mathsf{C}}(c, x) \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle f_*} \\ y & & \mathrm{Hom}_{\mathsf{C}}(c, y) \end{array}$$

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Remark 1.0.2. If we consider the Opposite Category C^{op} in Definition 1.0.13 for each $c \in \mathsf{C}^{\mathrm{op}}$ we get a Functor $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c, -) \to \mathsf{Set}$ defined as

$$\begin{array}{ccc} x & & \mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c,x) \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle f_*} \\ y & & \mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c,y) \end{array}$$

Therefore since $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c,x) = \mathrm{Hom}_{\mathsf{C}}(x,c)$ we get a functor $\mathrm{Hom}_{\mathsf{C}}(-,c) \to \mathsf{Set}$

$$\begin{array}{ccc} x & & \mathrm{Hom}_{\mathsf{C}}(x,c) \\ {\scriptstyle f}\downarrow & \longmapsto & \uparrow{\scriptstyle f^*} \\ y & & \mathrm{Hom}_{\mathsf{C}}(y,c) \end{array}$$

▽

Definition 1.0.14. A Functor $F : \mathsf{C} \to \mathsf{Set}$ is **representable** if there exists $c \in \mathsf{C}$ such that F is naturally isomorphic to the represented functor associated to c. ⌟

Definition 1.0.15 (Limit)**.** ⌟

Definition 1.0.16 (Colimit)**.** ⌟

Definition 1.0.17 (Equalizer)**.** ⌟

Definition 1.0.18 (Adjoint)**.** ⌟

Definition 1.0.19 (Product)**.** ⌟

Definition 1.0.20 (Coproduct)**.** ⌟

Definition 1.0.21. ⌟

Definition 1.0.22 (Subcategory)**.** ⌟

Definition 1.0.23 (Full Subcategory)**.** ⌟

Definition 1.0.24 (Directed Colimit)**.** ⌟

Proposition 1.0.1 (Functors preserve Isomorphisms)**.**

Definition 1.0.25 (Universal property)**.** ⌟

Lemma 1.0.1 (Yonneda lemma)**.**

Definition 1.0.26 (Algebraic Category)**.** ⌟